Koblitz cover transfers: Round25

An explicit original-source quotient and direct norms

2026-10-28 UTC. Exact algebra, native certificates and kernel proof records; no IC cost, new relation rank or official scientific promotion claim.

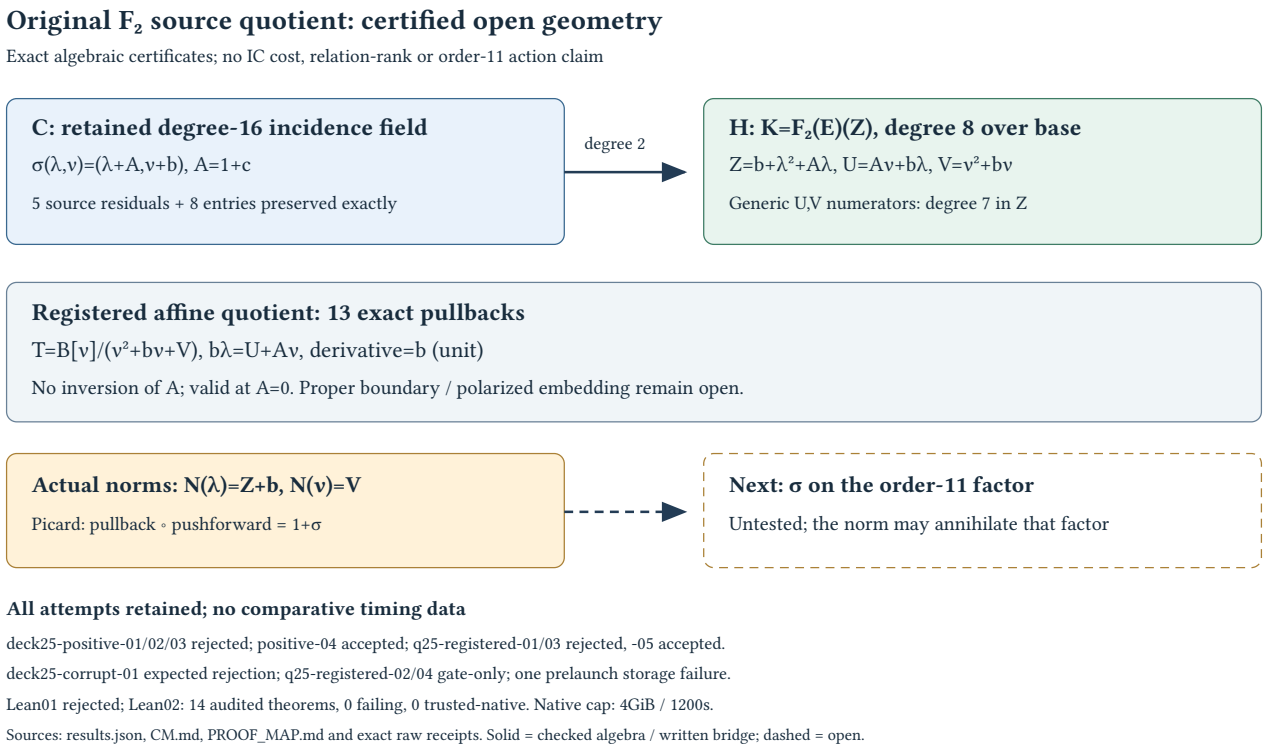
The actual quadratic action

In the unchanged degree16 incidence field L over $E : v^2 + v = u^3 + u$, put $A = 1 + c$. The action $\sigma(\lambda, \nu) = (\lambda + A, \nu + b)$ squares to identity and exactly preserves all five original equations, all eight registered entries and the literal generic nu equation. Its invariants are $Z = b + \lambda^2 + A\lambda$, $U = A\nu + b\lambda$, $V = \nu^2 + b\nu$. Native reduction gives U and V as degree7 polynomials in Z divided by $A\Delta$ and $A^2\Delta$, respectively. The actual retained quadratic extension identifies the fixed field as $K = F_2(E)(Z)$.

The integral chart retains A=0

Thirteen polynomials Q are explicitly extracted from the original five equations and eight entries, with exact pullbacks $Q(A\nu + b\lambda, \nu^2 + b\nu, b, c) = b^d P(\nu, \lambda, b, c)$. Their b-powers are0,0,6,7,8,0,0,0,0,0,0,3,2. Full coefficients are in registered-quotient.sing, copied unchanged from the accepted native output.

Let B be the quotient by Q1,...,Q5 localized at Q6,...,Q13. Since b=Q6 is a unit, the original registered source algebra is explicitly $T = B[\nu]/(\nu^2 + b\nu + V)$, with $b\lambda = U + A\nu$. This monic rank2 presentation has unit derivative b, hence is finite etale. Writing each element as p+q nu shows that the sigma-invariants are exactly B. No inversion of A is used, so the chart remains valid at A=0. These scheme consequences use the written ring-isomorphism argument in CM.md; the full proper boundary and proper polarized embedding remain open.



Norms, verification and remaining acceptance

The exact quadratic formulas are $\mathrm{Tr}(\lambda) = A$, $\mathrm{N}(\lambda) = Z + b$, $\mathrm{Tr}(\nu) = b$, $\mathrm{N}(\nu) = V$, and $A^2V = U^2 + AbU + b^2(Z + b)$. For p,q in K, expansion gives the written general formula $\mathrm{N}(p + q\nu) = p^2 + bpq + Vq^2$. It supplies a function norm once arithmetic on H is available. An arbitrary divisor-class reverse evaluator is still required.

A necessary subgroup test

The proper normalized degree2 map has $\pi^*\pi_* = 1 + \sigma$ on Picard groups under the written curve correspondence. Thus its norm can kill the desired order11 factor when sigma acts as minus1. The next acceptance check is an actual subgroup or matrix certificate for sigma and F2 Frobenius. The projector6(1+sigma) modulo11 would isolate a nonzero plus part if present; no such factor action or modular-Jacobian identification is inferred here.

All attempted work is recorded

Record class	Count	Outcome
Native invocations	8	2 accepted,5 implementation rejections,1 expected rejection
Gate-only refusals	2	No native input/output
Storage prelaunch failure	1	/tmp inode exhaustion; task-owned /var/tmp used
Lean attempts	2	1 rejected elaboration,1 accepted14-theorem audit

The corrupted actual invariant has residual1. Native exit0 and a later marker occur, but the wrapper rejects it as required. All accepted native blocks require zero residuals, native exit0, marker and no error diagnostics. The implementation failures are reserved identifiers or map syntax, not mathematical obstructions. Caps stay4GiB virtual/1200s. No migration, new EKS job, duplicate Diem, fifth task claim or PR merge occurs.

Lean4.34.1 / pinned mathlib audits14 new theorems,0 failing,0 trusted-native. Invariant, inverse, involution, trace/norm and mixed identities are formalized. The actual function-field, normal-curve, affine-scheme and Picard bridges are written, not kernel formalized; see PROOF_MAP.md. R24 separately records 14 audited theorems, four rational infinity branches e1,1,7,7 and actual coordinate/denominator valuations; its source and70-member archive are unchanged.

The integrated root library run hits1200s, exit124, with no full summary. The final27-test diagnostic passes26 and fails1 inherited registry contract (unknown ic.hyperelliptic). Exact fixture restoration changes no root code or expectation. This software evidence is separate from algebra/proofs and does not establish a green full suite or completed CI.

Open construction requirements

Resolve A0,B0,C0,D0 and enumerate additional zero supports; complete global divisors must precede a complete prescribed Riemann-Roch space and the F2 factor-map test. Node187 still needs source-bound membership/separator or a complete coupled arc. Full quotient saturation/proper polarized embedding, arbitrary reverse evaluation and ordinary460/367 images remain open. R22 integral augmentation recovery d=1 is preserved and not rerun.

Sources checked before approach: R22 generic/tower certificates,R23/R24 geometry and Stacks09I7, 03PA, 0BXX. Reproduce using PROTOCOL.md, REPLAY_NATIVE.sh, REPLAY_LEAN.sh and VERIFY_RETAINED.sh.